

UNIT-3 GRAPHS

Graph and Graph models - Graph terminology
and special types of graph - Matrix
Representation of graphs and graph Isomorphism
connectivity - Euler and Hamilton paths.

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1. (i) prove that the number of odd degree vertices is always even
(ii) state and prove the fundamental theorem of graph theory.

SOLUTION:

(i) Let V_1 and V_2 be the set of all vertices of even degree and the set of all vertices of odd degree, respectively, in the graph $G=(V,E)$

$$\therefore \sum d(v) = \sum_{v_i \in V_1} d(v_i) + \sum_{v_j \in V_2} d(v_j)$$

By Handshaking theorem, we have

$$2e = \sum_{v_i \in V_1} \deg(v_i) + \sum_{v_j \in V_2} \deg(v_j)$$

Since each $\deg(v_i)$ is even, $\sum_{v_i \in V_1} \deg(v_i)$ is even

$$\Rightarrow 2e = \text{an even number} + \sum_{v_j \in V_2} \deg(v_j)$$

$$\Rightarrow \sum_{v_j \in V_2} \deg(v_j) = (2e) - (\text{an even number})$$

$$\sum_{v_j \in V_2} \deg(v_j) = \text{an even number}$$

Since each term of $\deg(v_j)$ is odd.
Therefore number of term in the LHS
sum must be even.

$$\Rightarrow \sum_{v_j \in V_2} \deg(v_j) \text{ must be even}$$

(i.e.,) the number of vertices of odd
degree is even.

(ii) Fundamental theorem of Graph theory

Statement: For any finite undirected
graph G with vertex set V and edge set
 E , the sum of the degrees of all vertices
equals twice the number of edges:

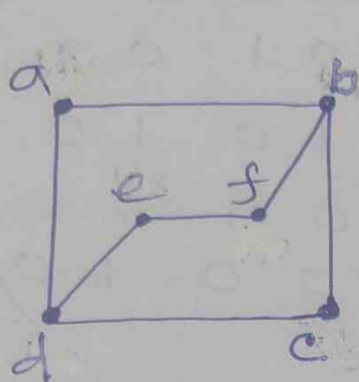
$$(i.e.,) \sum_{v \in V} \deg(v) = 2|E|$$

Proof: each edge in the graph contributes
exactly 1 to the degree of two distinct
vertices (its endpoints), so the total
sum of degrees counts each edge twice

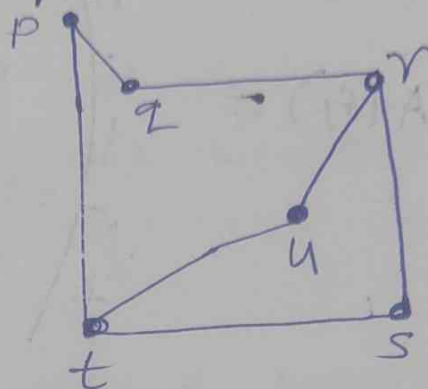
Thus $\sum_{v \in V} \deg(v) = 2|E|$, as both sides
enumerate the endpoints of all edges.

Therefore, the theorem is proved

2. Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem and check the graphs are isomorphism



Graph: G



Graph: H

PROOF:

- (i) Same number of vertices (6)
- (ii) Same number of edges (7)
- (iii) The corresponding vertices with same degree sequence

(i.e.) $G: 2 \ 2 \ 2 \ 2 \ 3 \ 3$
 $H: 2 \ 2 \ 2 \ 2 \ 3 \ 3$

(iv) define $f: G \rightarrow H$

$$\begin{array}{l|l} f(a) = u & f(d) = r \\ f(b) = t & f(e) = q \\ f(c) = s & f(f) = p \end{array}$$

Therefore f is one to one and onto corresponding between the vertices.

(V) adjacency matrix of the graph

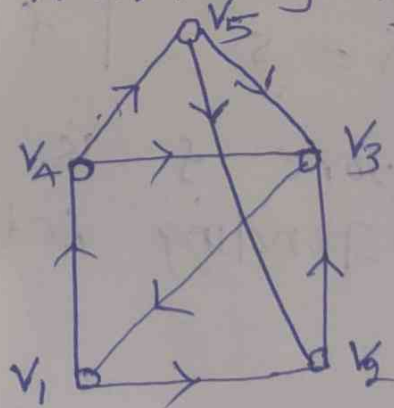
$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix} \end{pmatrix}$$

$$A(H) = \begin{matrix} & \begin{matrix} p & q & r & s & t & u \end{matrix} \\ \begin{matrix} p \\ q \\ r \\ s \\ t \\ u \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix} \end{pmatrix}$$

$$\Rightarrow A(G) = A(H)$$

Hence the Graphs are Isomorphism to each other.

3. (i) Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify the handshaking theorem.



(ii) state and prove Handshaking theorem
(or)

Let $G = (V, E)$ be an undirected graph with
'e' edges. Then $\sum_{v \in V} \deg(v) = 2e$.

SOLUTION :

(i)

$|V|$ = number of vertices = 5

$|E|$ = number of edges = 8

$\text{Indeg}(V_1) = 1$	$\text{outdeg}(V_1) = 2$	Total $\deg(V_1) = 3$
$\text{Indeg}(V_2) = 2$	$\text{outdeg}(V_2) = 1$	Total $\deg(V_2) = 3$
$\text{Indeg}(V_3) = 3$	$\text{outdeg}(V_3) = 1$	Total $\deg(V_3) = 4$
$\text{Indeg}(V_4) = 1$	$\text{outdeg}(V_4) = 2$	Total $\deg(V_4) = 3$
$\text{Indeg}(V_5) = 1$	$\text{outdeg}(V_5) = 2$	Total $\deg(V_5) = 3$

Min degree, $\delta(v) = 1$

Max degree, $\Delta(v) = 3$

using Handshaking theorem,

$$\sum_{i=1}^n d(v_i) = 2|E| = \sum \deg^+ + \sum \deg^-$$

$$\sum_{i=1}^n d(v_i) = 8 + 8 = 16 = 2 \times 8$$

where $e = 8$

(ii) statement:

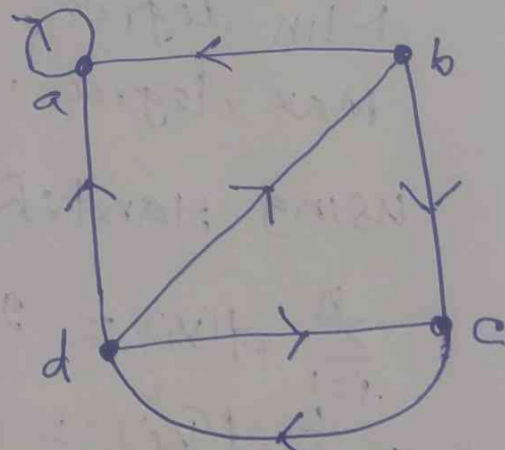
The sum of degrees of all the vertices of an undirected graph is twice the number of edges of the graph and hence even.

Proof:
Since every edge is incident with exactly two vertices, every edge contributes 2 to the sum of the degree of the vertices.

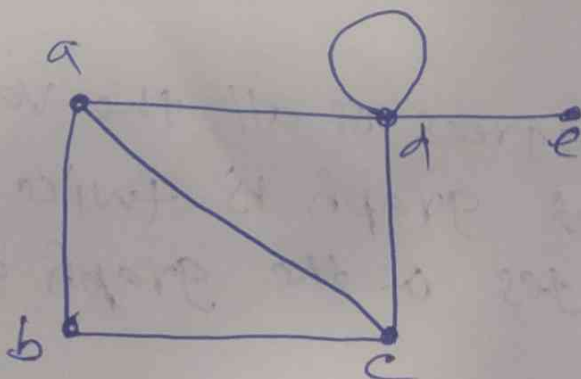
$\therefore$ All the "e" edges contribute $(2e)$ to the sum of the degree of vertices

$$\therefore \sum \deg(V) = 2|E| = 2e$$

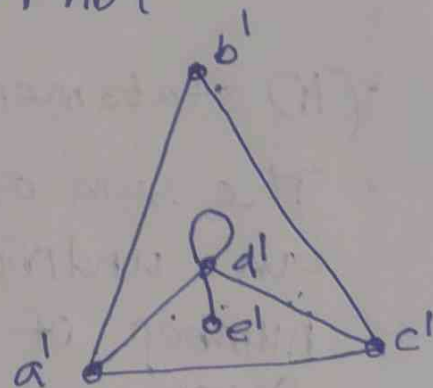
4. (i) Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem.



(ii) Exam whether the following pair of graphs are isomorphic or not



Graph: G



Graph: G'

SOLUTION:

- (i) (a) Number of Vertices, $|V| = 4$
(b) Number of edges, $|E| = 7$

(c)

$\text{Indeg}(a) = 3$	$\text{outdeg}(a) = 1$	$\text{Total deg}(a) = 4$
$\text{Indeg}(b) = 1$	$\text{outdeg}(b) = 2$	$\text{Total deg}(b) = 3$
$\text{Indeg}(c) = 2$	$\text{outdeg}(c) = 1$	$\text{Total deg}(c) = 3$
$\text{Indeg}(d) = 1$	$\text{outdeg}(d) = 3$	$\text{Total deg}(d) = 4$

- (d) Minimum degree, $\delta(v) = 1$
Maximum degree, $\Delta(v) = 3$

(e) using Handshaking theorem,

$$\sum_{i=1}^n d(v_i) = 2|E| = \sum \text{deg}^+ + \sum \text{deg}^-$$

//: deg^+ (indegree); deg^- (outdegree)

$$\Rightarrow \sum_{i=1}^n d(v_i) = 7 + 7 = 14 = 2 \times 7 = 2|E| = 2e$$

where $e = 7$

- (ii) (a) The number of Vertices, $|V| = 5$

(b) The number of Edges, $|E| = 8$

Hence the number of edges and Vertices of the graphs are same.

© The degree sequence of G : 3 2 3 5 1

The degree sequence of G' : 3 2 3 5 1

Hence the degree sequence are same.

④ define a function $f: G \rightarrow G'$

$$\left. \begin{array}{l} f(a) = a' \\ f(b) = b' \\ f(c) = c' \end{array} \right| \begin{array}{l} f(d) = d' \\ f(e) = e' \end{array}$$

Therefore f is a one to one and onto corresponding between the vertices and edges.

⑤ Adjacency matrices:

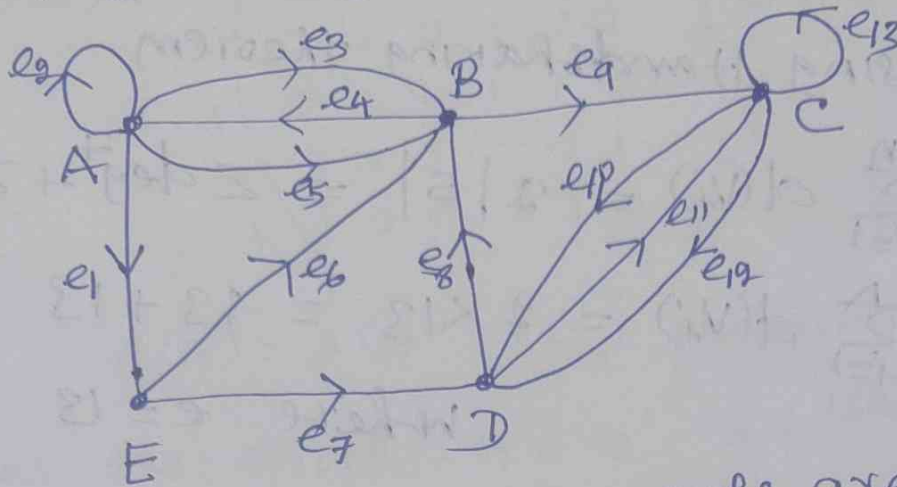
$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}$$

$$A(G') = \begin{matrix} & \begin{matrix} a' & b' & c' & d' & e' \end{matrix} \\ \begin{matrix} a' \\ b' \\ c' \\ d' \\ e' \end{matrix} & \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix} \Rightarrow A(G) = A(G')$$

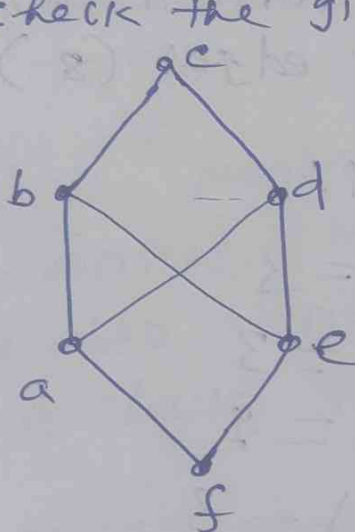
The adjacency is preserved

Thus G and G' are isomorphism to each other

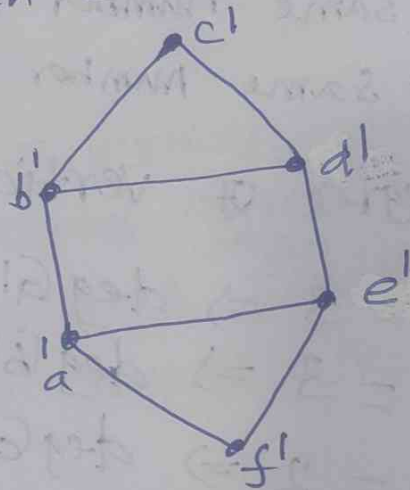
8. (i) Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with Handshaking theorem



- (ii) check the given two graphs are isomorphic?



Graph: G



Graph: G'

SOLUTION:

- (i) Number of vertices, $|V| = 5$
 Number of edges, $|E| = 13$

Graph	indegree deg+	outdegree deg-
a	2	4
b	4	2
c	3	3
d	3	2
e	1	2

Minimum degree, $\delta(V) = 1$

Maximum degree, $\Delta(V) = 4$

using Handshaking theorem

$$\sum_{i=1}^n d(V_i) = 2|E| = \sum \deg^+ + \sum \deg^-$$

$$\sum_{i=1}^n d(V_i) = 2 \times 13 = 13 + 13$$

where $e = 13$

(ii) (a) same number of vertices (6)

(b) same number of edges (8)

(c) degree of vertices:

$$d(a) = 3 \Rightarrow \deg(a) = 3$$

$$d(b) = 3 \Rightarrow \deg(b) = 3$$

$$d(c) = 2 \Rightarrow \deg(c) = 2$$

$$d(d) = 3 \Rightarrow \deg(d) = 3$$

$$d(e) = 3 \Rightarrow \deg(e) = 3$$

$$d(f) = 2 \Rightarrow \deg(f) = 2$$

(ie.,) same degree sequences $(3, 3, 2, 3, 3, 2)$

(d) define $f: G \rightarrow G'$

$$f(a) = a'$$

$$f(b) = b'$$

$$f(c) = c'$$

$$f(d) = d'$$

$$f(e) = e'$$

$$f(f) = f'$$

$$a \rightarrow a'$$

$$b \rightarrow b'$$

$$c \rightarrow c'$$

$$d \rightarrow d'$$

$$e \rightarrow e'$$

$$f \rightarrow f'$$

Therefore the function f is one to one and onto
corresponding between the vertices
(ie.,) There exists a bijection

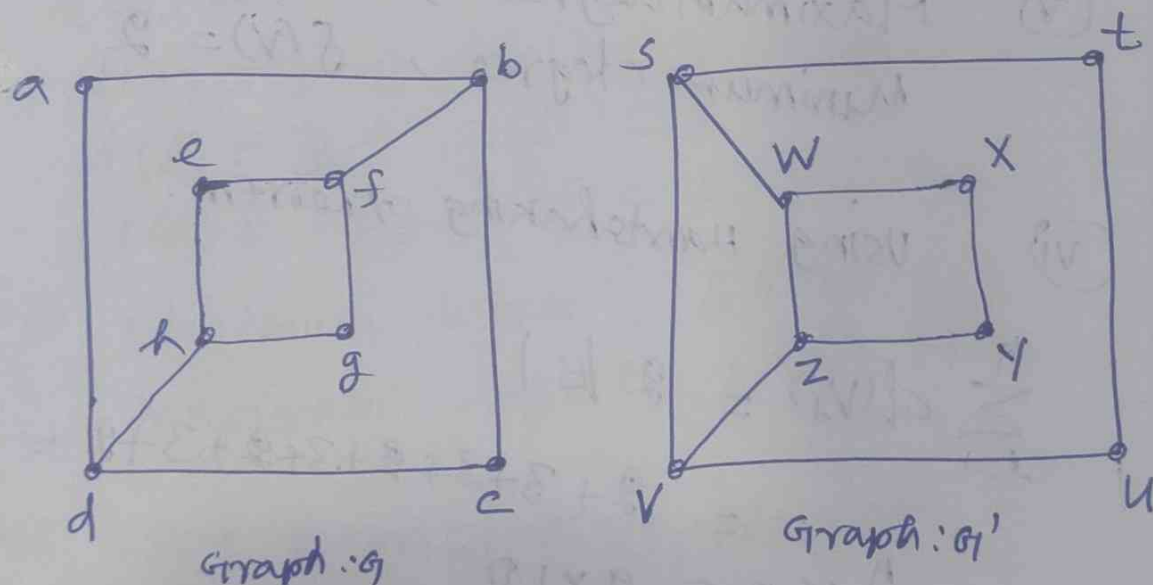
(e) Adjacency matrix of the graph:

$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix} = A(G')$$

$\Rightarrow A(G) = A(G')$ (i.e.) preserving adjacency

Thus the graphs G & G' are isomorphic to each other.

9. Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem and check the graphs are isomorphic.



SOLUTION:

(i) Number of vertices, $|V| = 8$

(ii) Number of Edges, $|E| = 10$

(iii) degree of vertices

$$d(a) = 2 \Rightarrow d(t) = 2$$

$$d(b) = 3 \Rightarrow d(s) = 3$$

$$d(c) = 2 \Rightarrow d(u) = 2$$

$$d(d) = 3 \Rightarrow d(v) = 3$$

$$d(e) = 2 \Rightarrow d(x) = 2$$

$$d(f) = 3 \Rightarrow d(w) = 3$$

$$d(g) = 2 \Rightarrow d(y) = 2$$

$$d(h) = 3 \Rightarrow d(z) = 3$$

(iv) define $f: G \rightarrow G'$

f is not one-to-one and onto
(the corresponding vertices with different degree
-sequences)
 $A(G) \neq A(G')$

(v) Maximum degree, $\Delta(V) = 3$
Minimum degree, $\delta(V) = 2$

(vi) Using Handshaking theorem.

$$\sum_{i=1}^n d(v_i) = 2|E|$$

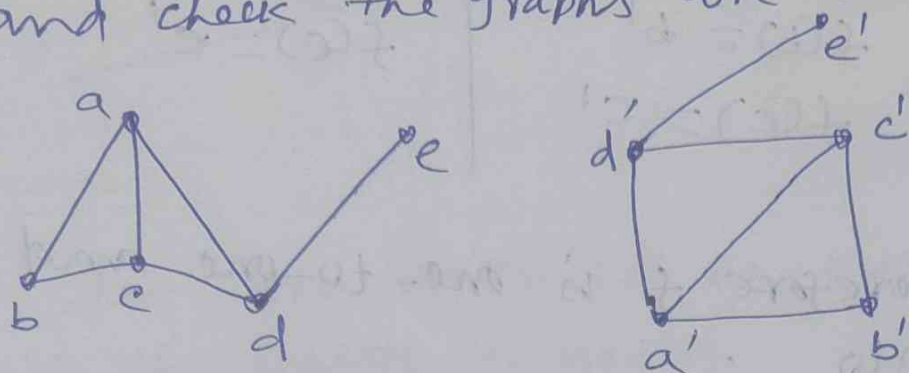
$$= 2 + 3 + 3 + 2 + 3 + 2 + 3 + 2 = 20$$

$$\sum_{i=1}^n d(v_i) = 2 \times 10$$

where $e = |E| = 10$

$\therefore$ Hence the graphs G & G' are not
Isomorphism to each other.

10. Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with Handshaking theorem and check the graphs are isomorphism



SOLUTION:

- (i) Number of vertices, $|V| = 5$
 (ii) Number of edges, $|E| = 6$
 (iii) Degree of vertices

$$d(a) = 3 \Rightarrow d(a') = 3$$

$$d(b) = 2 \Rightarrow d(b') = 2$$

$$d(c) = 3 \Rightarrow d(c') = 3$$

$$d(d) = 3 \Rightarrow d(d') = 3$$

$$d(e) = 1 \Rightarrow d(e') = 1$$

- (iv) Minimum degree, $\delta(V) = 1$
 Maximum degree, $\Delta(V) = 3$

- (v) using Handshaking theorem,

$$\sum_{i=1}^n d(v_i) = 2|E| = \sum \deg^+ + \sum \deg^-$$

$$\sum_{i=1}^5 d(v_i) = 3+2+3+3+1 = 12 = 2 \times 6$$

where $e = 6$.

(vi) def $f: G \rightarrow G'$

$$f(a) = a'$$

$$f(b) = b'$$

$$f(c) = c'$$

$$f(d) = d'$$

$$f(e) = e'$$

Therefore f is one-to-one and onto.

(vii) Adjacency matrices:

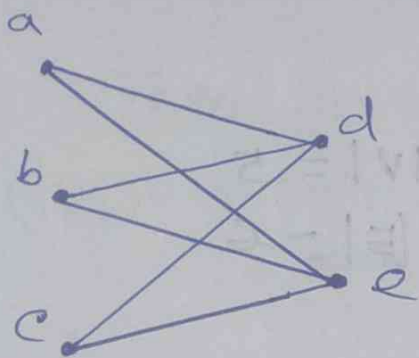
$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \end{matrix} = A(G')$$

$$\Rightarrow A(G) = A(G')$$

Hence the graphs G and G' are isomorphic to each other.

z. (i) For any simple bipartite graph G with edges and vertices prove that $e \leq \frac{v^2}{4}$

(ii) Find the number of vertices, edges, degrees of vertices and also verify with Handshaking theorem, maximum degree and minimum degree



SOLUTION:

① Let the Vertex set be partitioned into the subsets V_1 and V_2 .

Let V_1 contain x vertices. Then V_2 contains $(V-x)$ vertices.

The Largest number of edges of the graph can be obtained, when each of the x vertices in V_1 is connected to each of the $(V-x)$ vertices in V_2 .

∴ The largest number of edges,

$$f(x) = x(V-x) = (Vx - x^2)$$

$$f'(x) = (V - 2x)$$

$$f''(x) = (-2)$$

For stationary points $f'(x) = 0 \Rightarrow V - 2x = 0$

$$\Rightarrow \boxed{x = \frac{V}{2}}$$

(ie,) $f''(\frac{V}{2}) = (-2) < 0$, Hence $f(x)$ is maximum when $x = \frac{V}{2}$.

$$\therefore \text{Maximum number of edges required} \\ = f\left(\frac{V}{2}\right) = \left(V \cdot \frac{V}{2} - \left(\frac{V}{2}\right)^2\right) = \frac{V^2}{2} - \frac{V^2}{4} = \frac{V^2}{4}$$

(ii)

(a) Number of vertices, $|V| = 5$

(b) Number of edges, $|E| = 6$

(c) degree of vertices:

$$d(a) = 2$$

$$d(b) = 2$$

$$d(c) = 2$$

$$d(d) = 3$$

$$d(e) = 3$$

(d) using Handshaking theorem

$$\sum_{i=1}^n d(v_i) = 2|E|$$

$$\sum_{i=1}^n d(v_i) = 2 + 2 + 2 + 3 + 3 = 12$$

$$= 2 \times 6$$

where $e = 6$

(e) Maximum degree, $\Delta(V) = 3$

Minimum degree, $\delta(V) = 2$

6. Prove that, G is a simple graph with n vertices and k components when the number of edges at most

$$\frac{(n-k)(n-k+1)}{2}$$

SOLUTION:

let $n_1, n_2, \dots, n_k$ be the number of vertices in each of k components of the graph G .

Then $n_1 + n_2 + \dots + n_k = n = |V(G)|$

(i.e.,) $\sum_{i=1}^k n_i = n \rightarrow \textcircled{1}$

$$\begin{aligned} \sum_{i=1}^k (n_i - 1) &= (n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1) \\ &= (n_1 + n_2 + \dots + n_k) - (1 + 1 + \dots + 1)_{k \text{ times}} \end{aligned}$$

$$\sum_{i=1}^k n_i - 1 = n - k \rightarrow \textcircled{2}$$

Squaring on both sides

$$(n_1 - 1)^2 + (n_2 - 1)^2 + \dots + (n_k - 1)^2 \leq n^2 + k^2 - 2nk$$

$$n_1^2 - 2n_1 + 1 + n_2^2 - 2n_2 + 1 + \dots + n_k^2 - 2n_k + 1 \leq n^2 + k^2 - 2nk$$

$$\begin{aligned} (n_1^2 + n_2^2 + \dots + n_k^2) - 2(n_1 + n_2 + \dots + n_k) + (1 + 1 + \dots + 1)_{k \text{ times}} \\ \leq n^2 + k^2 - 2nk \end{aligned}$$

$$\Rightarrow \sum_{i=1}^k n_i^2 - 2n + k \leq n^2 + k^2 - 2nk$$

$$\Rightarrow \sum_{i=1}^k n_i^2 \leq (n^2 + k^2 - 2nk + 2n - k)$$

$$\Rightarrow \sum_{i=1}^k n_i^2 \leq (n^2 + k^2 - 2nk) + 2n - k \rightarrow \textcircled{3}$$

Since G is simple graph the maximum number of edges of G in its components

is $\frac{n_i (n_i - 1)}{2}$

∴ Maximum number of edges of G

$$= \sum_{i=1}^k \frac{n_i (n_i - 1)}{2}$$

$$= \frac{1}{2} \left\{ \left(\sum_{i=1}^k n_i^2 \right) - \left(\sum_{i=1}^k n_i \right) \right\}$$

$$= \frac{1}{2} \left\{ (n^2 + k^2 - 2nk) + 2n - k - n \right\}$$

$$= \frac{1}{2} \left\{ (n-k)^2 + (n-k) \right\}$$

$$= \frac{1}{2} (n-k)(n-k+1)$$

∴ Maximum number of edges

$$\leq \frac{(n-k)(n-k+1)}{2}$$

5. (i) prove that a given connected graphs G is a Euler graphs if all vertices of G is even degree.

(ii) Give an example for Eulerian but not Hamiltonian graph

(iii) Hamiltonian but not Eulerian graph

SOLUTION:

(i) Let G be an Eulerian graph.

We have to prove all vertices are of even degree.

Since G is Eulerian, G contains an Euler Circuit, say

$$v_0, e_1, v_1, e_2, \dots, v_{n-1}, e_n, v_0.$$

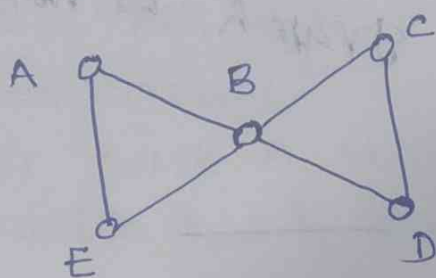
Both the edges e_1 and e_n contribute one to the degree of v_0 and so $\deg v_0$ is at least two.

In tracing this circuit we find an edge enters a vertex and another edge leaves the vertex contributing 2 to the degree of the vertex.

This is true for all vertices and so each vertex is of degree 2, an even integer.

$\therefore G$ has all the vertices of even degree.

(ii) Example of Eulerian graph but not a Hamiltonian graph is



Graph : G_1

G_1 contains the Eulerian cycle is

$A-B-C-D-B-E-A$

($\because$ All the edges occur exactly once).

Moreover the degree of all vertices of G_1 is even.

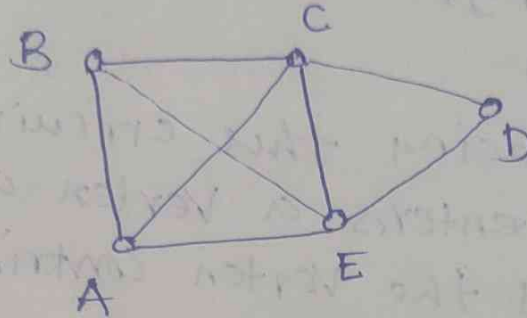
$\therefore G_1$ is Euler graph.

We cannot find Hamiltonian cycle as the vertex B is repeated twice.

$\therefore G_1$ is not a Hamiltonian graph.

Therefore G_1 is Eulerian but not Hamiltonian.

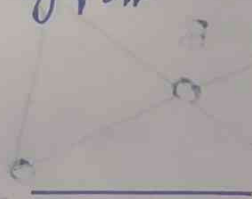
(iii) Example of Hamiltonian but not Eulerian is



Graph G_2

Since G_2 contains the Hamiltonian cycle, namely $A-B-C-D-E-A$ (All the vertices occurs exactly once).

G_2 is Hamiltonian graph
Since the degree of the vertex A is 3, $\deg(A)$ is not an even.
Therefore given graph is not an Euler graph.

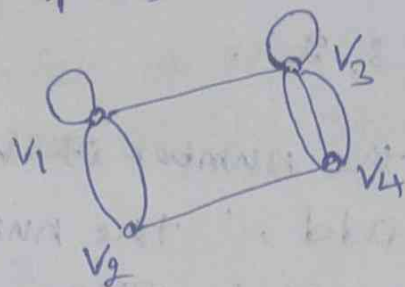


PART - A

1. Define pseudo graph and give example

A graph is said to be pseudo graph if it has both parallel edges and self loops.

For example,



2. can a simple graph be existing with 15 vertices each of degree is 5?

Let e be the number of edges of the graph

(ie.) $2e = \text{sum of all degrees}$

$$2e = 15 \times 5$$

$$2e = 75$$

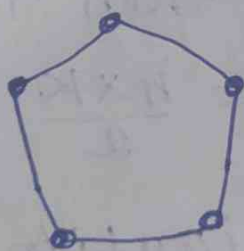
$$// \quad e = (75/2)$$

of degree 5 cannot exist, because the total degree is odd.

3. Define cyclic graph and draw the graph C_5 .

If every vertex of a simple graph G is of degree 2, then G is called a cycle or cyclic graph or circuit.

A cycle with n vertices is denoted by $C_n, n \geq 3$. Thus a cyclic graph is 2-regular.



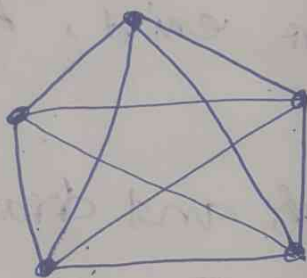
Graph: C_5

4. If there is a graph corresponding to a following degree $4, 4, 4, 3, 2$.

NO, because the number of vertices with odd degree is odd, the number of vertices of odd degree must be even.

5. Define complete graph and draw the graph K_5 .

A graph G is said to be complete if every vertex in G is connected to every other vertex in G . Thus a complete graph G must be connected.



Graph: K_5

6. What is the number of edges in K .
Regular graph.

If the graph has n vertices and it is k -regular, then the number of edges is $|E| = \frac{n \times k}{2}$.

7. Define a complete bipartite graph of $K_{2,3}$.

In a graph theory, a complete Bipartite graph $K_{m,n}$ is a graph where the

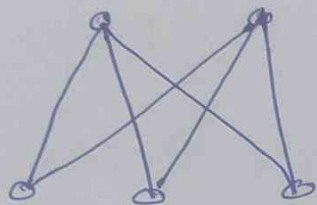
Vertices are divided into two sets, and every vertex of one set is connected to every vertex of the other set.

EXA: $K_{2,3}$

First set $M = \{m_1, m_2\}$

and second set $N = \{n_1, n_2, n_3\}$

Each vertex in set M connects to all vertices in set N



Graph: $K_{2,3}$

8. can be a regular graph be a complete graph.

A regular graph is a graph where every vertex has the same degree.

A complete graph is a graph where every vertex is connected to every other vertex.

Therefore, a complete graph is always a regular graph, but not every regular graph is complete.

9. Define incidence matrix and give an example

Let G be a graph with n vertices. Let

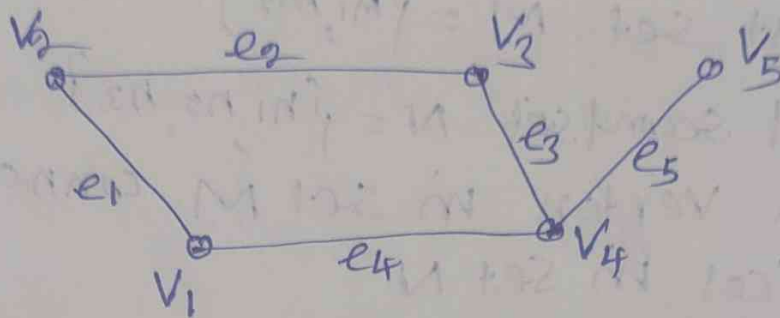
$V = \{v_1, v_2, \dots, v_n\}$ and $E = \{e_1, e_2, e_3, \dots, e_m\}$

Define $n \times m$ matrix

$$I_G = [m_{ij}]_{n \times m} \text{ where}$$

$$m_{ij} = \begin{cases} 1, & \text{when } v_i \text{ is incident with } e_j \\ 0, & \text{otherwise} \end{cases}$$

EXA:



$$I_G = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

10. What is the number of edges in complete graph, K_n

A complete graph K_n is a graph where every pair of distinct vertices is connected by an edge.

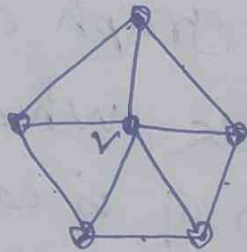
Number of edges in K_n

If the graph has n vertices, the number of edges is $|E| = \frac{n(n-1)}{2}$.

11. Define isomorphism of two graphs.

Two graphs are isomorphism if there exists a one-to-one mapping between their vertices that preserves adjacency.

12. How many vertices and edges in W_5 wheel graph W_n ($n \geq 3$) is obtained from C_n by adding a vertex v inside C_n and connecting v to every vertex of C_n by new edges.



Graph: W_5

(i.e.,)

number of vertices, $|V| = 5$

number of edges, $|E| = 10$

13. Define a pendent vertex in a graph

A vertex of a graph is called a pendant vertex if only one edge is incident with it.

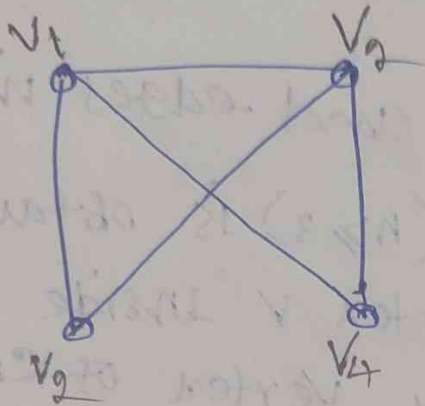
14. Define Adjacency matrix and give an Example

Let $G = \{V, E\}$ be a simple graph with n vertices ordered from V_1 to V_n , then the adjacency matrix $A = [a_{ij}]_{n \times n}$ of G is an $n \times n$ symmetric matrix defined by the elements

$$a_{ij} = \begin{cases} 1, & \text{when } v_i \text{ is adjacent to } v_j \\ 0, & \text{otherwise} \end{cases}$$

It is denoted by $A(G)$ or A_G

EXA:

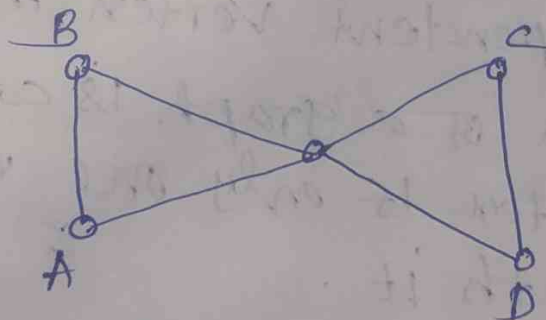


$$A_G = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix} \end{matrix}$$

15. Define Euler graph and given an example.

A closed walk which contains all edges of the graph G is called an Euler line, and the graph containing at least one Euler line is said to be an Euler graph.

EXA:



Consider the cycle $A-B-C-D-A$

Since, it includes each of the edges exactly once, the above cycle is a Eulerian cycle. Since the graph contains Eulerian cycle, it is Euler graph.